

PROBLEM SET 2

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score: 75**)

1) **(15 pts)** For the matrix below:

(a) **(10 pts)** Find the eigenvalues and normalized eigenvectors of the matrix

$$\Omega = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 2 & 0 \\ 0 & 1 & 4 \end{bmatrix}$$

(b) **(3 pts)** Is the matrix Hermitian ?

(c) **(2 pts)** Are the eigenvectors orthogonal ?

2) **(15 pts)** Consider the matrix

$$\Omega = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

(a) **(3 pts)** Is the matrix Hermitian ?

(b) **(10 pts)** Find the eigenvalues and eigenvectors

(c) **(2 pts)** Verify that $U^\dagger \Omega U$ is diagonal,
(U being the matrix of eigenvectors of Ω)

3) **(15 pts)** Consider the Hermitian matrix

$$\Omega = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

(a) **(5 pts)** Show that the eigenvalues are $\omega_1 = \omega_2 = 1; \omega_3 = 2$

(b) **(5 pts)** Show that $|\omega = 2\rangle$ is any vector of the form

$$\frac{1}{\sqrt{2a^2}} \begin{bmatrix} 0 \\ a \\ -a \end{bmatrix}$$

(c) **(5 pts)** Show that the $\omega = 1$ eigenspace contains all vectors of the form

$$\frac{1}{\sqrt{b^2 + 2c^2}} \begin{bmatrix} b \\ c \\ c \end{bmatrix}$$

either by feeding $\omega = 1$ into the equations or by requiring that the $\omega = 1$ eigenspace be orthogonal to $|\omega = 2\rangle$.

- 4) **(10 pts)** An arbitrary $n \times n$ matrix need not have n eigenvectors. Consider as an example

$$\Omega = \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$$

- (a) **(5 pts)** Show that $\omega_1 = \omega_2 = 3$

- (b) **(5 pts)** By feeding in this value show we get only one eigenvector of the form

$$\frac{1}{\sqrt{2a^2}} \begin{bmatrix} +a \\ -a \end{bmatrix}$$

5) **(20 pts)** Consider the matrix

$$\Omega = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

(a) **(3 pts)** Show that it is unitary

(b) **(10 pts)** Show that its eigenvalues are $e^{i\theta}$ and $e^{-i\theta}$

(c) **(5 pts)** Find the corresponding eigenvectors; show that they are orthogonal

(d) **(2 pts)** Verify that $U^\dagger \Omega U = (\text{diagonal matrix})$, where U is the matrix of eigenvectors of Ω